

UNIFORM SHEAFY TATE DOMAINS THAT ARE NOT STABLY UNIFORM

ANONYMOUS

ABSTRACT. Work of Buzzard–Verberkmoes and, independently, Mihara shows that stable uniformity is a sufficient criterion for a Tate Huber ring to be sheafy, meaning that the structure presheaf of topological rings on its associated adic space is a sheaf. In this paper we construct examples of uniform sheafy rings which are not stably uniform, answering Question 7 of Kedlaya’s *Nonarchimedean Scottish Book*. The examples were found in collaboration with ChatGPT 5.6 Sol. The first has been fully formalised in Lean 4; its proof rests on showing that a particular strict pullback of complete Tate rings is preserved by rational localisation. Formalisation of the second is ongoing: the construction, uniformity, sheafiness, and the nonuniform domain chart have been checked, but reducedness of every finite iterated rational localisation has not.

1. INTRODUCTION

The point at issue is the behaviour of rational localisations. The examples of Buzzard–Verberkmoes and Mihara show that a uniform Tate Huber ring can lose uniformity on a rational chart and, separately, that uniformity alone does not imply sheafiness [5, 15]. They leave open whether sheafiness itself prevents this loss of uniformity. Hansen and Kedlaya ask this explicitly [7, Remark 3.16]; it is also Question 7 of Kedlaya’s *Nonarchimedean Scottish Book* [16]. The answer is no.

We begin with the finite-jet pullback. This is a nonnoetherian uniform integral domain for which a distinguished rational localisation is nonreduced, and hence is no longer uniform. Let k be a complete discretely valued nonarchimedean field, with valuation ring k° , uniformizer ϖ , and residue field $\bar{k}$. Put

$$\begin{aligned} L_0 &= k^\circ\langle W, W^{-1} \rangle, & B_0 &= k^\circ\langle W, Q \rangle/(Q^2), \\ C_0 &= L_0\langle Q \rangle, & D_0 &= L_0\langle Q \rangle/(Q^2), \end{aligned}$$

where $k^\circ\langle W, W^{-1} \rangle$ means $k^\circ\langle W, V \rangle/(WV - 1)$. There are natural maps $B_0 \rightarrow D_0$ and $C_0 \twoheadrightarrow D_0$. Define

$$A_0 = B_0 \times_{D_0} C_0, \quad A = A_0[1/\varpi],$$

and endow A with the topology defined by the ring of definition A_0 .

The main result is the following.

Theorem 1.1. *The ring A is a complete uniform nonnoetherian Tate k -algebra and an integral domain, with $A^\circ = A_0$. The Huber pair (A, A°) is sheafy. However,*

$$A\left\langle \frac{W}{\varpi} \right\rangle \cong k\langle X, Q \rangle/(Q^2), \quad X = \frac{W}{\varpi},$$

so A is not stably uniform.

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The proof uses Milnor squares of complete Tate rings. Recall that a cartesian square of rings with a surjective leg is called a Milnor square [14, Section 2]; the complete Tate setting is used by Kerz–Saito–Tamme in their work on analytic K -theory [11, Section 3].

Definition 1.2. A commutative square of complete Tate k -algebras

$$\begin{array}{ccc} R & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & D \end{array}$$

is a *strict Milnor square* if it is cartesian, the sequence of underlying topological k -vector spaces

$$0 \longrightarrow R \longrightarrow B \oplus C \xrightarrow{(b,c) \mapsto \bar{b} - \bar{c}} D \longrightarrow 0$$

is strict exact, and $C \rightarrow D$ is a strict surjection.

After rotating the square if necessary, the initial Milnor square is strict by the nonarchimedean open mapping theorem [11, Lemma 3.1]. Our main input is that strictness persists after completed rational localisation and is compatible with refinement. This allows sheafiness to be transferred from B, C, D to R . The associated square of affinoid adic spaces is a pushout among affinoid spaces, but not among all adic spaces; see remark 6.3. The second example shows that the failure of stable uniformity need not come from a nonreduced localisation.

Both examples were found in collaboration with ChatGPT 5.6 Sol. The first has been formalised in Lean, and substantial parts of the second have also been checked. Appendix A gives the precise scope of the two developments and relates their statements to the definitions used here.

Remark 1.3 (Relation with earlier work). Buzzard–Verberkmoes construct a uniform affinoid ring with a nonuniform rational localisation and, separately, a uniform affinoid ring whose structure presheaf is not a sheaf [5, Propositions 17 and 18]. Mihara gives further examples of both phenomena [15]. These examples do not give a uniform sheafy ring which is not stably uniform.

The strict Koszul result used in lemma 5.1 is due to Ben-Bassat–Kremnizer and Bambozzi–Kremnizer [2, Definition 5.7 and Lemmas 5.13–5.14] [3, Definitions 4.5–4.6, Proposition 4.9, Lemma 4.11, and Corollary 4.13]. We use its lattice form to prove that rational localisation preserves the finite-jet Milnor square. The resulting uniform sheafy domain which is not stably uniform appears to be new.

The weighted-parity construction of section 7 is closer to the infinite monomial-support examples of Buzzard–Verberkmoes and Mihara. Its additional feature is that rational localisation can be calculated in a noetherian finite head.

2. CONVENTIONS AND NOTATION

We follow Huber for Huber pairs, adic spectra, rational subsets, and rational localisation [8], and use Wedhorn’s account for strong noetherianity and sheafiness [18, Proposition and Definition 6.36, Section 8.1]; the bounded-denominator language for strictness follows Buzzard–Verberkmoes [5], and the quotient presentation of a rational localisation is also discussed by Hansen–Kedlaya [7].

- *Tate topology.* Throughout, E denotes a complete Tate k -algebra. In the discretely valued setting of this paper, this means that E contains a topologically nilpotent unit and admits an open bounded k° -subalgebra E_0 such that

$$E = E_0[1/\varpi], \quad \{\varpi^n E_0\}_{n \geq 0}$$

is a basis of neighbourhoods of zero. Such an E_0 is called a *ring of definition*. We always choose it closed, hence ϖ -adically complete. A subset of E is bounded if it is contained in $\varpi^{-r}E_0$ for some $r \geq 0$. If M is a complete topological k -vector space, an open bounded k° -submodule $M_0 \subset M$ will be called a lattice.

- *Uniformity and noetherianity.* We write E° for the ring of power-bounded elements. The ring E is *uniform* if E° is bounded, and *strongly noetherian* if $E\langle Z_1, \dots, Z_s \rangle$ is noetherian for every $s \geq 0$. A Tate ring is *stably uniform* if every rational localisation is uniform.
- *Huber pairs and sheafiness.* A *ring of integral elements* is an open subring $E^+ \subseteq E^\circ$ which is integrally closed in E . We call the Huber pair (E, E^+) *sheafy* when its structure presheaf is a sheaf of complete topological rings; in particular, the map to the matching-family equaliser for a finite rational cover is a homeomorphism.
- *Strictness.* We use the following bounded-denominator form of strictness. Let $u : M \rightarrow N$ be a continuous k -linear map and choose lattices $M_0 \subset M$ and $N_0 \subset N$. Then u is strict if and only if there exists $a \geq 0$ such that

$$(1) \quad \varpi^a(u(M) \cap N_0) \subset u(M_0).$$

For a surjection this becomes $\varpi^a N_0 \subset u(M_0)$. For an injection it says that the lattice induced from the target is bounded in the source. This is the criterion used by Buzzard–Verberkmoes [5, Lemma 2]. We use the term *strict Milnor square* in the sense of definition 1.2; the underlying algebraic terminology is standard [14, 11].

- *Rational localisation.* We take every rational datum to have at least one numerator; an empty list may be padded by $f_1 = 0$. Let $\alpha = (f_1, \dots, f_m; g)$ be a rational datum in E , so the ideal generated by $g, f_1, \dots, f_m$ is open. Since E is a Tate k -algebra, this ideal is the unit ideal. After multiplying the entire datum by a common power of ϖ , we may assume that all entries lie in E_0 .

The rational localisation E_α is the separated ϖ -adic completion of $E[1/g]$ for the ring of definition generated by

$$E_0 \left[\frac{f_1}{g}, \dots, \frac{f_m}{g} \right].$$

It is independent of the ring of definition and of the presentation of the rational domain, and it is transitive under rational refinement. Put

$$T_E = E\langle T_1, \dots, T_m \rangle = E_0\langle T_1, \dots, T_m \rangle[1/\varpi], \quad r_i = gT_i - f_i,$$

and

$$(2) \quad I_{E,\alpha} := (r_1, \dots, r_m) = \text{im} \left(T_E^m \xrightarrow{d_{1,E}} T_E \right) \subset T_E, \quad d_{1,E}(u_1, \dots, u_m) = \sum_i u_i(gT_i - f_i),$$

Then

$$(3) \quad E_\alpha \cong T_E / \overline{I_{E,\alpha}}.$$

Here the bar is necessary until the ideal has been shown to be closed. This is the standard quotient presentation of rational localisation; see [8, (1.2), Proposition 1.3, Lemma 1.5] and [7, Definition 3.6].

3. THE GLOBAL RING

In this section we identify the finite-jet algebra with a concrete subring of C and prove that it is a complete uniform domain. Recall that

$$L_0 = k^\circ\langle W, W^{-1} \rangle, \quad B_0 = k^\circ\langle W, Q \rangle / (Q^2),$$

$$C_0 = L_0\langle Q \rangle, \quad D_0 = L_0\langle Q \rangle / (Q^2),$$

where $k^\circ\langle W, W^{-1} \rangle$ denotes $k^\circ\langle W, V \rangle / (WV - 1)$. The natural maps $B_0 \rightarrow D_0$ and $C_0 \rightarrow D_0$ define

$$A_0 = B_0 \times_{D_0} C_0.$$

We write

$$A = A_0[1/\varpi], \quad L = L_0[1/\varpi], \quad B = B_0[1/\varpi], \quad C = C_0[1/\varpi], \quad D = D_0[1/\varpi].$$

Each of these Tate rings carries the topology defined by its subscript-zero ring.

The map $C_0 \rightarrow D_0$ has a continuous k° -linear section given by truncation in the Q -variable:

$$f_0 + Qf_1 \mapsto f_0 + Qf_1.$$

Consequently

$$(4) \quad 0 \longrightarrow A_0 \longrightarrow B_0 \oplus C_0 \longrightarrow D_0 \longrightarrow 0$$

is exact, and A_0 is ϖ -adically complete. After inverting ϖ we obtain a strict Milnor square

$$\begin{array}{ccc} A & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & D. \end{array}$$

In fact, the chosen rings of definition already make the integral row exact, so no powers of ϖ are lost in the defining square.

The map $B_0 \rightarrow D_0$ is injective. Projection to C_0 therefore identifies A_0 with the subring

$$(5) \quad A_0 = \{f_0(W) + Qf_1(W) + Q^2h(W, W^{-1}, Q) : f_0, f_1 \in k^\circ\langle W \rangle, h \in C_0\}.$$

Equivalently, A_0 consists of the restricted series

$$(6) \quad \sum_{(a,b) \in S} c_{a,b} W^a Q^b, \quad S = \{(a,b) \in \mathbf{Z} \times \mathbf{N} : b \leq 1 \Rightarrow a \geq 0\},$$

with $c_{a,b} \in k^\circ$. Here restricted means that, for every n , only finitely many coefficients are nonzero modulo ϖ^n . The set S is closed under addition, so this is a subring of C_0 . The same description with coefficients in k gives $A \subset C$.

Equip A, B, C, D with the coefficientwise Gauss norms whose unit balls are the displayed rings of definition. Write

$$\iota_B : A \rightarrow B, \quad \iota_C : A \rightarrow C, \quad \rho_B : B \rightarrow D, \quad \rho_C : C \rightarrow D$$

for the structural maps, and give A the restricted Gauss norm from C . For $a = f_0 + Qf_1 + Q^2h$, the coefficient formulas give

$$(7) \quad \|\iota_C(a)\|_C = \|a\|_A, \quad \|\iota_B(a)\|_B \leq \|a\|_A, \quad \|\rho_B(b)\|_D \leq \|b\|_B, \quad \|\rho_C(c)\|_D \leq \|c\|_C.$$

Thus every structural map is norm-nonincreasing.

Proposition 3.1. *The ring A is a complete uniform nonnoetherian Tate k -algebra and an integral domain. Moreover,*

$$A^\circ = A_0.$$

Proof. The Gauss norm on $L = k\langle W, W^{-1} \rangle$ is multiplicative, and the restricted Gauss norm on $C = L\langle Q \rangle$ is therefore multiplicative as well. In particular, C is a domain, and hence so is its subring A . The conditions on the constant and linear Q -coefficients which define A are closed conditions in C , so A is complete. The image of ϖ in A is a topologically nilpotent unit, and therefore A is a Tate ring.

The coefficient description of A_0 shows that it is exactly the unit ball for the restricted Gauss norm on A . Since this norm is multiplicative,

$$\|a^n\| = \|a\|^n.$$

It follows that a is power-bounded if and only if $\|a\| \leq 1$. Thus $A^\circ = A_0$, and the unit ball is bounded, so A is uniform.

For nonnoetherianity, let $J = \ker(\iota_B) = Q^2C \subset A$. If J were generated by $a_1, \dots, a_r$, then, for every $\ell \in L$, comparison of the Q^2 -coefficient in an expression

$$Q^2\ell = \sum_i x_i a_i, \quad x_i \in A,$$

would express ℓ as a $k\langle W \rangle$ -linear combination of the Q^2 -coefficients of the a_i . This would make $L = k\langle W, W^{-1} \rangle$ a finite $k\langle W \rangle$ -module, which is impossible: a finite algebra is integral, whereas a monic equation for W^{-1} , after multiplication by a power of W , would put 1 in the proper ideal $(W) \subset k\langle W \rangle$. Thus J is not finitely generated, and A is not noetherian. $\square$

4. A NONUNIFORM RATIONAL LOCALISATION

We next compute the rational chart which witnesses the failure of stable uniformity.

Consider the datum

$$\alpha_{\text{in}} = (W; \varpi).$$

The ideal (ϖ, W) is open, since ϖ is a unit in A . Let

$$A_{\text{in}} = A \left\langle \frac{W}{\varpi} \right\rangle$$

be the corresponding rational localisation on $|W| \leq |\varpi|$, and write $X = W/\varpi$. By the ring-of-definition construction, A_{in} is the separated completion of the subring generated by A_0 and X .

Proposition 4.1. *There is a canonical isomorphism of complete topological rings*

$$A_{\text{in}} \cong k\langle X, Q \rangle / (Q^2), \quad W \mapsto \varpi X, \quad Q \mapsto Q.$$

Proof. Put

$$T = k\langle X, Q \rangle / (Q^2).$$

Take the two-jet map $A \rightarrow B$ and then rescale its disc variable by $W \mapsto \varpi X$. This gives a continuous map

$$\theta : A \longrightarrow T, \quad f_0(W) + Qf_1(W) + Q^2h \longmapsto f_0(\varpi X) + Qf_1(\varpi X).$$

The element $\theta(\varpi)$ is a unit and $\theta(W)/\theta(\varpi) = X$ is power-bounded. The universal property first extends θ to the algebraic rational localisation and then, by continuity, to a map

$$\Phi : A_{\text{in}} \longrightarrow T.$$

We next record why the two-jet map loses no information other than the Q^2 -tail after completion. If the constant and linear Q -coefficients of $y \in A$ vanish, then $y_n = W^{-n}y$ still belongs to A , and the y_n have bounded norm. In A_{in} one has

$$\bar{y} = \varpi^n X^n \overline{y_n} \quad (n \geq 0).$$

The powers of X are bounded and $\varpi^n \rightarrow 0$, so $\bar{y} = 0$. In particular, the image $\bar{Q}$ of Q satisfies $\bar{Q}^2 = 0$.

Let $\bar{X} = \bar{W}/\varpi$. Since $\bar{X}$ is power-bounded, evaluation at $\bar{X}$ gives a continuous map $k\langle X \rangle \rightarrow A_{\text{in}}$. We may therefore define

$$\Psi : T \longrightarrow A_{\text{in}}, \quad f + Qg \longmapsto f(\bar{X}) + g(\bar{X})\bar{Q}.$$

To check the first composite, decompose $a \in A$ as $a = f_0(W) + Qf_1(W) + y$, where the first two coefficients lie in $k\langle W \rangle$ and the first two Q -coefficients of y vanish. The preceding paragraph kills y , and hence $\Psi(\theta(a)) = \bar{a}$. This proves $\Psi\Phi = 1$ on the dense algebraic rational localisation, and therefore on A_{in} . In the other direction, $\Phi\Psi$ fixes constants, X and Q . It is consequently the identity first on polynomials, then on $k\langle X \rangle$ by density, and finally on T . Thus Φ and Ψ are inverse continuous homomorphisms. $\square$

Proposition 4.2. *The ring A is not stably uniform.*

Proof. In $T = k\langle X, Q \rangle / (Q^2)$, every element of the line kQ is nilpotent and hence power-bounded. This line is not bounded with respect to the ring of definition T_0 : for every $r \geq 0$,

$$\varpi^r(\varpi^{-(r+1)}Q) = \varpi^{-1}Q \notin T_0.$$

Thus T is not uniform. By proposition 4.1, a rational localisation of the uniform ring A is nonuniform. $\square$

5. LOCALISATION OF THE MILNOR SQUARE

In this section we prove that rational localisation preserves the defining pullback. We first record the strict Koszul estimate needed for the proof.

5.1. Koszul complexes for rational localisations. Let E be one of B, C, D , with its coefficientwise Gauss norm and the displayed unit ball E_0 . Let $f_1, \dots, f_m, g \in E_0$ generate the unit ideal in E . Put

$$T_E = E\langle T_1, \dots, T_m \rangle, \quad T_{E,0} = E_0\langle T_1, \dots, T_m \rangle,$$

and set $r_i = gT_i - f_i$.

We give T_E the topology for which $\{\varpi^n T_{E,0}\}_{n \geq 0}$ is a basis of neighbourhoods of zero. The term $\bigwedge^q T_E^m$ has the finite-module topology defined by the lattice $\bigwedge^q T_{E,0}^m$; equivalently, in the standard exterior basis it has the finite product topology. Images have the subspace topology. Thus a differential is strict when it is open onto its image, or equivalently when its coimage is homeomorphic to its image.

The following result is a special case of the strict Koszul statements in [2, Definition 5.7 and Lemmas 5.13–5.14] [3, Definitions 4.5–4.6, Proposition 4.9, Lemma 4.11, and Corollary 4.13]. We include the argument used below, since the topology and the two losses of denominators are important.

Lemma 5.1. *The Koszul complex*

$$0 \longrightarrow \bigwedge^m T_E^m \longrightarrow \cdots \longrightarrow \bigwedge^2 T_E^m \xrightarrow{d_{2,E}} T_E^m \xrightarrow{d_{1,E}} T_E$$

is exact in positive degrees. Every differential is continuous and strict onto its image, and every image is closed. In particular, if

$$I_E = (r_1, \dots, r_m) = \text{im}(d_{1,E}) \subset T_E, \quad d_{1,E}(u_1, \dots, u_m) = \sum_i u_i r_i,$$

then I_E is closed. Moreover, there exists $h_E \in \mathbf{Z}_{\geq 0}$ such that

$$(8) \quad \varpi^{h_E}(I_E \cap T_{E,0}) \subset d_{1,E}(T_{E,0}^m).$$

There also exists $z_E \in \mathbf{Z}_{\geq 0}$ such that

$$(9) \quad \varpi^{z_E}(\ker(d_{1,E}) \cap T_{E,0}^m) \subset d_{2,E}(\bigwedge^2 T_{E,0}^m).$$

Proof. Over the polynomial ring $E[T_1, \dots, T_m]$, the sequence $gT_i - f_i$ has Koszul complex exact in positive degrees. The map

$$E[T_1, \dots, T_m] \longrightarrow T_E$$

is flat: this is the usual identification of T_E with the ϖ -adic completion of $E_0[T_1, \dots, T_m]$, followed by inverting ϖ . Tensoring the polynomial Koszul complex with T_E , and using the standard base-change identification of its finite free terms, proves positive-degree exactness. Notice that this does not assert exactness in degree zero.

Each differential is continuous because its coordinates are finite sums of multiplication maps. The image of $d_{1,E}$ is closed by the noetherian closed-ideal theorem. In higher degree, exactness identifies an image with the kernel of the next continuous differential, and hence that image is closed. The nonarchimedean open mapping theorem, applied to each differential with its closed image, now gives a constant C_q such that every element of the image has a preimage of norm at most C_q times its norm. This proves strictness.

For $d_{1,E}$ choose h_E so that $|\varpi|^{h_E} C \leq 1$, where C is a controlled-lift constant for $d_{1,E}$. Scaling a controlled lift of an element of $I_E \cap T_{E,0}$ gives (8). Applying the same argument to $d_{2,E}$, with its own constant and using $\text{im}(d_{2,E}) = \ker(d_{1,E})$, gives (9). $\square$

5.2. The defining ideals under pullback. Fix a rational datum

$$\alpha = (f_1, \dots, f_m; g)$$

in A . Multiplying all entries by a common power of ϖ , we assume $f_1, \dots, f_m, g \in A_0$. For $E \in \{A, B, C, D\}$ use the notation T_E , $T_{E,0}$, and $d_{1,E}$ introduced above, and write

$$I_E = (gT_1 - f_1, \dots, gT_m - f_m) \subset T_E.$$

The exact integral row (4) and the coefficientwise section of $C_0 \rightarrow D_0$ give an exact sequence

$$(10) \quad 0 \longrightarrow T_{A,0} \longrightarrow T_{B,0} \oplus T_{C,0} \longrightarrow T_{D,0} \longrightarrow 0.$$

The map $T_{C,0} \rightarrow T_{D,0}$ is surjective, as are the induced maps on finite free modules and their exterior powers. Exactness is coefficientwise: a compatible restricted coefficient pair lifts to A_0 , and the lifted coefficients still tend to zero ϖ -adically.

Lemma 5.2. *The ideal I_A is closed in T_A , and the canonical map*

$$I_A \longrightarrow I_B \times_{I_D} I_C$$

is an isomorphism. Moreover,

$$(11) \quad 0 \longrightarrow I_A \longrightarrow I_B \oplus I_C \longrightarrow I_D \longrightarrow 0$$

is strict exact.

Proof. We use the controlled lifts supplied in the proof of lemma 5.1. Let $(x_B, x_C) \in I_B \times_{I_D} I_C$, and choose lifts $u_B \in T_B^m$ and $u_C \in T_C^m$ of x_B and x_C whose norms are bounded by fixed multiples of $\|x_B\|$ and $\|x_C\|$. In T_D^m the difference

$$w = (u_B)_D - (u_C)_D$$

belongs to $\ker(d_{1,D})$. Choose a controlled lift $v_D \in \bigwedge^2 T_D^m$ with $d_{2,D}(v_D) = w$, and lift v_D coefficientwise to $v_C \in \bigwedge^2 T_C^m$. The coefficientwise section $T_C \rightarrow T_D$ preserves the norm. Replacing u_C by

$$u'_C = u_C + d_{2,C}(v_C)$$

does not change $d_{1,C}(u_C)$, and the images of u_B and u'_C in T_D^m now agree. Exactness of (10), applied coefficientwise, gives $u_A \in T_A^m$ mapping to this pair. Then $d_{1,A}(u_A)$ maps to (x_B, x_C) .

The same construction keeps track of the norms. The controlled lifts for $d_{1,B}$, $d_{1,C}$ and $d_{2,D}$, continuity of $d_{2,C}$, and the norm-one coefficientwise section give a constant C , independent of (x_B, x_C) , for which the lift $x_A = d_{1,A}(u_A)$ may be chosen with

$$\|x_A\| \leq C \max\{\|x_B\|, \|x_C\|\}.$$

The map $T_A \rightarrow T_B \oplus T_C$ is injective, so this proves both the algebraic pullback assertion and the boundedness of its inverse.

It remains to see that I_A is closed. The ideals I_B and I_C are closed by lemma 5.1. If x lies in the closure of I_A , its images lie in I_B and I_C . The controlled pullback just proved gives $x_A \in I_A$ with the same two images as x . Injectivity of $T_A \rightarrow T_B \oplus T_C$ gives $x = x_A$, as required.

Finally, let $y \in I_D$. A controlled lift of y through $d_{1,D}$, followed by the coefficientwise section $T_D^m \rightarrow T_C^m$, produces $x_C \in I_C$ mapping to y , with norm bounded by a fixed multiple of $\|y\|$. Thus $I_B \oplus I_C \rightarrow I_D$ is a strict surjection. Its kernel is I_A by the pullback assertion, and the preceding norm estimate makes the inclusion on the left strict. This proves (11). $\square$

Proposition 5.3. *For every rational datum α in A , the canonical sequence*

$$(12) \quad 0 \longrightarrow A_\alpha \longrightarrow B_\alpha \oplus C_\alpha \longrightarrow D_\alpha \longrightarrow 0$$

is strict exact, and $C_\alpha \rightarrow D_\alpha$ is a strict surjection. Equivalently,

$$A_\alpha \cong B_\alpha \times_{D_\alpha} C_\alpha.$$

The coefficient maps used in these identifications commute with rational restriction.

Proof. The ideals I_B, I_C, I_D are closed by lemma 5.1, and I_A is closed by lemma 5.2. Hence (3) gives

$$E_\alpha = T_E / I_E \quad (E = A, B, C, D).$$

We first prove algebraic exactness. If a representative $p \in T_A$ maps into both I_B and I_C , the controlled pullback in lemma 5.2 gives an element of I_A with the same two images. Injectivity of $T_A \rightarrow T_B \oplus T_C$ then shows that $p \in I_A$. This gives injectivity on the left. Conversely, take compatible classes in T_B/I_B and T_C/I_C and choose representatives b and c . Their defect in T_D belongs to I_D . Lift this defect to an element of I_C using the strict surjection in (11), and add it to c . The corrected representatives agree in T_D , so (10) lifts them to T_A . This proves (12).

For the topology on the left, let $x \in A_\alpha$ and put

$$M = \max\{\|x_B\|, \|x_C\|\}.$$

Choose representatives $b \in T_B$ and $c \in T_C$ whose norms are arbitrarily close to the quotient norms of x_B and x_C . Their defect lies in I_D . The controlled surjection $I_C \rightarrow I_D$ gives a correction $y \in I_C$ whose norm is bounded by a fixed multiple of the norm of this defect. Thus b and $c + y$ agree in T_D and lift to $p \in T_A$. The max-norm identity for the ambient pullback gives, after letting the error in the representatives tend to zero,

$$\|x\| \leq CM$$

for a constant C depending only on the datum. The two quotient maps are norm-nonincreasing, so this estimate says precisely that $A_\alpha \rightarrow B_\alpha \oplus C_\alpha$ is a topological embedding.

The coefficientwise section $T_D \rightarrow T_C$ shows directly that $C_\alpha \rightarrow D_\alpha$ is surjective. It also shows that $T_C \rightarrow T_D$ is open; passing to quotient seminorms proves that the induced map is open. Together with continuity, this makes $C_\alpha \rightarrow D_\alpha$ a strict surjection.

All maps used above come from the coefficient maps $A \rightarrow B, C$ and $B, C \rightarrow D$. More generally, a homomorphism $E \rightarrow E'$ carrying a rational datum α to α' induces a coefficientwise map $T_E \rightarrow T_{E'}$ which sends the relations for α to those for α' . The pullback identifications therefore commute with these maps. Applying this to rational restriction, and then using presentation independence and transitivity, proves the asserted compatibility under refinement. $\square$

6. SHEAFINESS

We isolate the elementary descent statement used to transfer sheafiness from the three comparison rings.

Lemma 6.1 (Sheaf transfer). *Let*

$$\begin{array}{ccc} R & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & D \end{array}$$

be a commutative square of complete Tate Huber pairs. For $E \in \{B, C, D\}$, write

$$p_E : X_E = \mathrm{Spa}(E, E^+) \longrightarrow X_R = \mathrm{Spa}(R, R^+)$$

for the induced map. For a rational domain $U \subset X_R$, put $U_E = p_E^{-1}(U)$, and set $U_R = U$. Suppose that for every rational domain $U \subset X_R$ the section rings satisfy a natural strict exact sequence

$$(13) \quad 0 \longrightarrow \mathcal{O}_R(U) \longrightarrow \mathcal{O}_B(U_B) \oplus \mathcal{O}_C(U_C) \longrightarrow \mathcal{O}_D(U_D) \longrightarrow 0.$$

If the Huber pairs associated with B, C, D are sheafy as complete topological rings, then the Huber pair associated with R is sheafy as a complete topological ring.

Proof. We first work on the rational basis. Let $U = \bigcup_{i=1}^n U_i$ be a finite rational covering. For $E \in \{R, B, C, D\}$, let

$$r_E : \mathcal{O}_E(U_E) \longrightarrow \prod_i \mathcal{O}_E((U_i)_E)$$

denote product restriction, using $U_R = U$ and $(U_i)_R = U_i$ as specified in the statement. The first map in (13) is a topological embedding, and the same is true after replacing U by any U_i . Since B and C are sheafy, r_B and r_C are topological embeddings. Naturality gives a commutative square

$$\begin{array}{ccc} \mathcal{O}_R(U) & \longrightarrow & \mathcal{O}_B(U_B) \oplus \mathcal{O}_C(U_C) \\ \downarrow r_R & & \downarrow r_B \oplus r_C \\ \prod_i \mathcal{O}_R(U_i) & \longrightarrow & \prod_i (\mathcal{O}_B((U_i)_B) \oplus \mathcal{O}_C((U_i)_C)). \end{array}$$

The top and right arrows are topological embeddings, as is the bottom arrow. It follows that r_R is a topological embedding.

Now let $x_i \in \mathcal{O}_R(U_i)$ be a matching family. Its images in the B - and C -section rings glue to elements

$$b \in \mathcal{O}_B(U_B), \quad c \in \mathcal{O}_C(U_C).$$

Their images in $\mathcal{O}_D(U_D)$ agree after restriction to every $(U_i)_D$. Since the structure presheaf for D is separated, they agree globally. Exactness on U therefore gives $x \in \mathcal{O}_R(U)$ mapping to (b, c) . Exactness on each U_i shows that $x|_{U_i} = x_i$, and uniqueness follows in the same way. Thus

r_R identifies $\mathcal{O}_R(U)$ homeomorphically with the matching-family equalizer, which is closed in the finite product because all section rings are Hausdorff.

The underlying presheaf is therefore a sheaf by the sheaf-on-a-basis theorem [17, Tag 009O]. To retain the topology on arbitrary opens, let $V = \bigcup_\lambda V_\lambda$ be an open covering and let M be its matching-family equalizer with the subspace topology. For every rational $U \subset V$, quasi-compactness gives a finite rational cover of U subordinate to the V_λ . Restriction from M to that finite cover, followed by the rational-basis result, gives a continuous map $M \rightarrow \mathcal{O}_R(U)$, independent of the chosen subordinate cover. These maps are compatible under rational restriction. Since

$$\mathcal{O}_R(V) = \varprojlim_{U \subset V, U \text{ rational}} \mathcal{O}_R(U)$$

with its projective-limit topology, the inverse of the algebraic gluing map $\mathcal{O}_R(V) \rightarrow M$ is continuous. Hence the gluing map is a homeomorphism, as required. $\square$

Theorem 6.2. *The Huber pair (A, A°) is sheafy.*

Proof. Give B, C, D their maximal rings of integral elements. The max/Gauss estimates (7) show that every structural map is norm-nonincreasing. Consequently it sends bounded subsets to bounded subsets; in particular, if x is power-bounded, then the set of powers of its image is the image of the bounded set $\{x^n : n \geq 0\}$. The structural maps therefore send power-bounded elements to power-bounded elements and define morphisms of Huber pairs.

Each of B, C, D is an affinoid k -algebra and is therefore sheafy by Huber's theorem [8, Theorem 2.2]. We apply the argument of lemma 6.1 directly to a finite rational cover $U = \bigcup_i U_i$ in $\text{Spa}(A, A^\circ)$. Push the datum for U and the data for the U_i to B, C, D . If a section over U restricts to zero on every U_i , its images over B and C vanish by separatedness there. Injectivity of the localised Milnor row from proposition 5.3 then shows that the original section is zero.

Now take a matching family over the U_i . Its images glue over B and C . The two glued sections have the same image over D , since their restrictions agree on the pushed cover and the presheaf for D is separated. Exactness of the localised Milnor row over U gives a section over A , and injectivity of the corresponding rows over the U_i shows that it restricts to the given family.

It remains to retain the topology. The image of the product restriction map is the matching-family equaliser just described, which is closed in the finite product of the section rings. The restriction map is continuous and injective, so the nonarchimedean closed-range theorem, using the topologically nilpotent unit of A , makes it a topological embedding. This proves the sheaf condition on finite rational covers. The sheaf-on-a-rational-basis theorem then gives the all-open topological sheaf condition for (A, A°) . $\square$

Remark 6.3 (Affinoid versus global pushouts). The coefficient descriptions of the maximal plus rings give

$$A^\circ = B^\circ \times_{D^\circ} C^\circ.$$

Indeed, a compatible pair which is power-bounded in B and C has its constant and linear Q -coefficients in both $k\langle W \rangle$ and L_0 ; their intersection is $k^\circ\langle W \rangle$, while the higher Q -tail is already integral in C_0 . Together with the pullback topology, this says that the defining square is a pullback of complete Huber pairs.

Put $X_E = \text{Spa}(E, E^\circ)$ for $E \in \{A, B, C, D\}$. Huber's affinoid mapping theorem identifies morphisms between affinoid adic spaces with continuous morphisms of the corresponding complete Huber pairs [8, Proposition 2.1]. Hence, for every affinoid adic space Z , the natural map

$$\text{Hom}(X_A, Z) \longrightarrow \text{Hom}(X_B, Z) \times_{\text{Hom}(X_D, Z)} \text{Hom}(X_C, Z)$$

is bijective. Thus the contravariant square

$$\begin{array}{ccc} X_D & \longrightarrow & X_C \\ \downarrow & & \downarrow \\ X_B & \longrightarrow & X_A \end{array}$$

is a pushout in the full subcategory of affinoid adic spaces.

This does not make it a pushout in the category of all adic spaces: there the same map must be bijective for an arbitrary, possibly nonaffinoid, target Z . In fact, uniqueness already fails for an explicit nonaffinoid target. Let

$$U_1 = \{x \in X_A : |Q^2(x)| \leq |\varpi(x)|\}, \quad U_2 = \{x \in X_A : |1| \leq |W(x)|\}, \quad U = U_1 \cup U_2.$$

The map $X_B \rightarrow X_A$ factors through U_1 , because $Q^2 = 0$ in B , while $X_C \rightarrow X_A$ factors through U_2 , because both W and W^{-1} are power-bounded in C and hence $|W| = 1$ on X_C .

The open subset U is proper. Use additive valuations with values in $\Gamma = \mathbf{Z}_{\text{lex}}^3$ and, for a nonzero restricted series written as in (6), put

$$v\left(\sum c_{a,b} W^a Q^b\right) = \min_{\text{lex}}\{(v_k(c_{a,b}), b, a) : c_{a,b} \neq 0\}.$$

Restrictedness makes the minimum exist, and the initial-term argument for the lexicographic monomial order makes v multiplicative. The condition $b \leq 1 \Rightarrow a \geq 0$ defining S makes it nonnegative on $A_0 = A^\circ$, while $v(\varpi) > 0$, so it is continuous and defines a point $x \in X_A$. Since

$$v(Q^2) = (0, 2, 0) < (1, 0, 0) = v(\varpi), \quad v(W) = (0, 0, 1) > 0,$$

one has $|Q^2(x)| > |\varpi(x)|$ and $|W(x)| < 1$; hence $x \notin U$.

Adic spaces may be glued along open subspaces. Glue two copies of X_A along U and write $Z = X_A^{(1)} \cup_U X_A^{(2)}$. The two open immersions $j_1, j_2 : X_A \rightarrow Z$ agree on U , so they agree after precomposition with both $X_B \rightarrow X_A$ and $X_C \rightarrow X_A$, but they are distinct at the two unglued copies of x . The displayed Hom map is therefore not injective for this Z . Thus X_A is not the pushout in the full category of adic spaces. More generally, the same doubling argument shows that a cocone whose two legs factor through a proper open subspace of its vertex cannot be a pushout in that category. We do not know whether the original span admits some other, necessarily nonaffinoid, pushout object.

Proof of theorem 1.1. Completeness, uniformity, nonnoetherianity, and the domain property are proposition 3.1; sheafiness is theorem 6.2; and failure of stable uniformity is proposition 4.2. $\square$

7. A SECOND EXAMPLE

The finite-jet construction above loses uniformity because separated completion creates a nilpotent. We now give a second construction in which the bad rational localisation is an integral domain. The loss of uniformity is instead caused by an unbounded family of nonnilpotent power-bounded elements.

Theorem 7.1. *There is a complete Tate k -algebra $\mathcal{A}$ with the following properties.*

- (1) $\mathcal{A}$ is a uniform, nonnoetherian integral domain and $\mathcal{A}^\circ = A_0$ for the ring of definition displayed below.
- (2) The Huber pair $(\mathcal{A}, \mathcal{A}^\circ)$ is sheafy.
- (3) Every finite iterated rational localisation of $\mathcal{A}$ is reduced.
- (4) The rational localisation

$$\mathcal{B} = \mathcal{A} \left\langle \frac{W}{\varpi} \right\rangle$$

is an integral domain but is not uniform.

In particular, failure of stable uniformity need not be caused by a nilpotent in the bad rational localisation.

7.1. The weighted-parity algebra. Let $I = \mathbf{N}_{>0}$. For a finite-support multi-index $\nu = (\nu_n)_{n \in I} \in \mathbf{N}^{(I)}$, put

$$(14) \quad \omega(\nu) = \sum_{\substack{n \geq 1 \\ \nu_n \text{ odd}}} n.$$

Consider the submonoid

$$(15) \quad S = \left\{ (a, \nu) \in \mathbf{N} \times \mathbf{N}^{(I)} : a \geq \omega(\nu) \right\}.$$

The inequality

$$(16) \quad \omega(\nu + \mu) \leq \omega(\nu) + \omega(\mu)$$

shows that S is closed under addition. It is locally finite: a fixed element of S has only finitely many decompositions as a sum of two elements of S .

Let $k[S]$ be the monoid algebra with basis $W^a U^\nu$, where $U^\nu = \prod_n U_n^{\nu_n}$. Give it the Gauss norm

$$(17) \quad \left\| \sum_{(a, \nu) \in S} c_{a, \nu} W^a U^\nu \right\|_G = \sup_{a, \nu} |c_{a, \nu}|.$$

Its completion is

$$(18) \quad \mathcal{A} = \left\{ \sum_{(a, \nu) \in S} c_{a, \nu} W^a U^\nu : c_{a, \nu} \in k, \quad c_{a, \nu} \longrightarrow 0 \right\},$$

where convergence to zero is over the discrete index set S . Local finiteness makes the coefficientwise convolution finite, and the submultiplicativity of the Gauss norm on $k[S]$ extends multiplication continuously to $\mathcal{A}$. Put

$$(19) \quad \mathcal{A}_0 = \left\{ \sum_{(a, \nu) \in S} c_{a, \nu} W^a U^\nu \in \mathcal{A} : c_{a, \nu} \in k^\circ \right\}.$$

Then $\mathcal{A}_0$ is the closed unit ball for (17), it is ϖ -adically complete, and $\mathcal{A} = \mathcal{A}_0[1/\varpi]$.

The finite stages admit a useful affinoid presentation. For $N \geq 1$, put

$$(20) \quad \mathcal{A}_N = \frac{k\langle W, Y_1, Z_1, \dots, Y_N, Z_N \rangle}{(Y_n^2 - W^{2n} Z_n)_{1 \leq n \leq N}}.$$

Lemma 7.2 (Isometric finite-stage normal form). *There is an isometric decomposition*

$$(21) \quad \mathcal{A}_N \cong \bigoplus_{\epsilon \in \{0,1\}^N}^{\max} k\langle W, Z_1, \dots, Z_N \rangle Y^\epsilon.$$

Under the substitution

$$(22) \quad Y_n \longmapsto W^n U_n, \quad Z_n \longmapsto U_n^2,$$

this identifies $\mathcal{A}_N$ isometrically with the closed subalgebra of $k\langle W, U_1, \dots, U_N \rangle$ supported on the monomials in (15) involving only $U_1, \dots, U_N$. The transition maps $\mathcal{A}_N \rightarrow \mathcal{A}_{N+1}$ are isometric, and

$$(23) \quad \mathcal{A} = \widehat{\bigcup_{N \geq 1} \mathcal{A}_N}.$$

Proof. Writing $\nu_n = 2q_n + \epsilon_n$ gives the unique factorization

$$(24) \quad W^a U^\nu = W^{a - \omega(\nu)} \prod_{n=1}^N Y_n^{\epsilon_n} Z_n^{q_n}$$

whenever $a \geq \omega(\nu)$ and ν is supported in $\{1, \dots, N\}$. Thus the finite-head support algebra is the sum of the subspaces

$$k\langle W, Z_1, \dots, Z_N \rangle Y^\epsilon, \quad \epsilon \in \{0, 1\}^N.$$

The summands have disjoint monomial supports, distinguished by the parities of the U_n -exponents. The sum is therefore direct and its Gauss norm is the maximum of the norms of its components.

The substitution (22) kills $Y_n^2 - W^{2n} Z_n$ and hence gives a norm-nonincreasing map from the quotient (20) to this support algebra. In the other direction, send a parity decomposition $\sum_\epsilon f_\epsilon Y^\epsilon$ to the class represented by the same sum. This map is also norm-nonincreasing, and (24) shows that the two maps are inverse. This proves the isometric normal form. The same support description makes the transition maps isometric. Finally, every finite set of allowed monomials lies in one head, and finite-support series are dense in (18); hence (23) follows. $\square$

Proposition 7.3. *The ring $\mathcal{A}$ is a complete uniform Tate k -algebra and an integral domain, with*

$$\mathcal{A}^\circ = \mathcal{A}_0.$$

Proof. The Gauss norm on the countable restricted Tate algebra $k\langle W, U_1, U_2, \dots \rangle$ is multiplicative. To see this, note first that the norm of a nonzero restricted series is attained. For each of two such series choose, among the norm-attaining exponents, the lexicographically largest one. In the coefficient of the sum of these two exponents, the product of the chosen coefficients strictly dominates every other convolution term. The ultrametric inequality therefore gives

$$\|fg\|_G = \|f\|_G \|g\|_G.$$

This is the countable restricted-series argument used in the formalisation, whose underlying Lean implementation is adapted from work of William Coram [6]. The support algebra $\mathcal{A}$ is a closed isometric subring of this Tate algebra, so its Gauss norm is multiplicative and it is a domain.

If $x \in \mathcal{A}_0$, all powers of x remain in $\mathcal{A}_0$. If $x \notin \mathcal{A}_0$, multiplicativity gives $\|x^m\|_G = \|x\|_G^m$, which is unbounded. Hence $\mathcal{A}^\circ = \mathcal{A}_0$. Completeness and the Tate property follow from (18) and the topologically nilpotent unit ϖ . $\square$

Proposition 7.4. *The ring $\mathcal{A}$ is not noetherian.*

Proof. For $m \geq 1$, let

$$I_m = (Z_1, \dots, Z_m) \subset \mathcal{A}.$$

Define a norm-nonincreasing homomorphism

$$\psi_m : \mathcal{A} \longrightarrow k\langle T \rangle,$$

by retaining only the coefficients of the monomials U_{m+1}^{2q} and sending that monomial to T^q . The support condition shows directly that this is multiplicative: a product can contribute such a monomial only when both factors contribute monomials of the same form. Thus ψ_m sends W and every Y_j to zero, sends Z_{m+1} to T , and sends every other Z_j to zero. It kills I_m but not Z_{m+1} . Therefore

$$I_1 \subsetneq I_2 \subsetneq I_3 \subsetneq \cdots,$$

so $\mathcal{A}$ is not noetherian. $\square$

7.2. A reduced nonuniform rational chart. The ideal (W, ϖ) is open because ϖ is a unit of $\mathcal{A}$. Define the complete k° -algebra

$$(25) \quad \mathcal{B}_0 = \left\{ \sum_{\nu \in \mathbf{N}^{(I)}, d \geq \omega(\nu)} b_{d,\nu} X^d U^\nu : b_{d,\nu} \in \varpi^{\omega(\nu)} k^\circ, \quad \varpi^{-\omega(\nu)} b_{d,\nu} \longrightarrow 0 \right\},$$

with norm

$$(26) \quad \left\| \sum b_{d,\nu} X^d U^\nu \right\|_{\text{in}} = \sup_{d,\nu} |\varpi|^{-\omega(\nu)} |b_{d,\nu}|.$$

The subadditivity (16) shows that $\mathcal{B}_0$ is a ring. Put $\mathcal{B} = \mathcal{B}_0[1/\varpi]$.

Proposition 7.5. *There is a canonical isometric isomorphism*

$$(27) \quad \mathcal{A} \left\langle \frac{W}{\varpi} \right\rangle \xrightarrow{\sim} \mathcal{B}, \quad W \mapsto \varpi X.$$

Proof. The datum $(W; \varpi)$ already lies in the zeroth head $H_0 = k\langle W \rangle$. Let

$$Q = H_0 \langle T \rangle / \overline{(\varpi T - W)}$$

be its separated completed quotient. Evaluation at $W = \varpi X$ and $T = X$ gives a continuous map $Q \rightarrow k\langle X \rangle$. Conversely, evaluation of $k\langle X \rangle$ at the image of T gives a continuous map in the other direction. The two maps are inverse by polynomial density, and the norm estimates in the two universal properties show that the isomorphism is isometric.

Apply the coefficientwise-localisation construction of proposition 7.8 to this zeroth head. It identifies the rational chart with the c_0 tail algebra over Q . After replacing Q by $k\langle X \rangle$, its elements and norm are exactly those in (25) and (26): the tail basis element of parity ν contributes the factor $\varpi^{\omega(\nu)}$. This gives (27), and the head evaluation sends W to ϖX . $\square$

Proposition 7.6. *The ring $\mathcal{B}$ is an integral domain but is not uniform.*

Proof. Group a series in $\mathcal{B}$ by its U -multi-index. This gives a coefficientwise homomorphism

$$\mathcal{B} \longrightarrow k\langle X \rangle[[U_1, U_2, \dots]].$$

It is injective, and the target is a domain because $k\langle X \rangle$ is a domain. Hence $\mathcal{B}$ is a domain. In the finite-head formalisation below, this is the embedding of the corresponding c_0 tail algebra into formal multivariable power series; the twisting element is ϖX , which is a non-zero-divisor.

Inside $\mathcal{A}$ write $Y_n = W^n U_n$, as in (22), and put

$$T_n = \varpi^{-n} Y_n = X^n U_n \in \mathcal{B}.$$

Formula (26) gives

$$\|T_n\|_{\text{in}} = |\varpi|^{-n}, \quad \|T_n^2\|_{\text{in}} = 1.$$

It follows by separating even and odd powers that the powers of each fixed T_n are bounded. Thus every T_n is power-bounded. The family $(T_n)_{n \geq 1}$ is not bounded, since its norms $|\varpi|^{-n}$ are unbounded. Hence $\mathcal{B}^\circ$ is unbounded and $\mathcal{B}$ is not uniform. $\square$

7.3. Small perturbations of rational data. The finite-head argument rests on the following elementary stability result.

Lemma 7.7 (Small perturbation lemma). *Let (E, E^+) be a complete Tate Huber pair and choose a closed ring of definition $E_0 \subseteq E^+$. Let $\alpha = (f_1, \dots, f_m; g)$ be a rational datum whose entries lie in E_0 . Write $d_0 = g$ and $d_i = f_i$ for $i \geq 1$. Suppose that for some $\ell \geq 0$ there are $a_0, \dots, a_m \in E_0$ with*

$$(28) \quad \varpi^\ell = \sum_{j=0}^m a_j d_j.$$

Let $d'_j \in E_0$ satisfy

$$(29) \quad d'_j - d_j \in \varpi^{\ell+1} E_0 \quad (0 \leq j \leq m),$$

and write $\alpha' = (f'_1, \dots, f'_m; g')$. Then α' is a rational datum, α and α' define the same rational subset of $\mathrm{Spa}(E, E^+)$, and their completed rational localisations are canonically isomorphic.

Proof. Write $d'_j - d_j = \varpi^{\ell+1} h_j$ with $h_j \in E_0$. Then

$$\sum_j a_j d'_j = \varpi^\ell (1 + \varpi H), \quad H = \sum_j a_j h_j \in E_0.$$

The element $1 + \varpi H$ is a unit of E_0 , with inverse given by its convergent geometric series. Multiplying the a_j by this inverse gives an integral Bezout relation for the primed datum; in particular, α' is open.

Let x lie in the rational subset defined by α . Since $a_j \in E_0 \subseteq E^+$, relation (28) gives

$$|\varpi|^\ell \leq |g(x)|.$$

The perturbations have value at most $|\varpi|^{\ell+1} < |g(x)|$, so $|g'(x)| = |g(x)|$ and $|f'_i(x)| \leq |g'(x)|$. This proves one inclusion of rational subsets. Applying the same argument to the primed Bezout relation proves the reverse inclusion.

Rational localisation is independent of the datum used to present a rational subset. The equality of the two subsets therefore gives restriction maps in both directions between E_α and $E_{\alpha'}$. Transitivity of rational restriction shows that their composites are the identity. This is the required canonical isomorphism. $\square$

7.4. Finite-head rational localisation. Fix $N \geq 1$. For a tail multi-index $\mu = (\mu_n)_{n>N} \in \mathbf{N}^{\{n>N\}}$, write $\mu_n = 2q_n + \epsilon_n$ with $\epsilon_n \in \{0, 1\}$ and put

$$(30) \quad e_\mu = \prod_{n>N} Y_n^{\epsilon_n} Z_n^{q_n}.$$

The finite-stage normal forms give an isometric Banach-module decomposition

$$(31) \quad \mathcal{A} \cong \widehat{\bigoplus_\mu}^{c_0} \mathcal{A}_N e_\mu.$$

Multiplication is determined by

$$(32) \quad e_\mu e_\lambda = W^{\omega(\mu) + \omega(\lambda) - \omega(\mu + \lambda)} e_{\mu + \lambda}.$$

Here ω is applied only to the tail coordinates. Projection to the coefficient of e_0 is a norm-nonincreasing algebra retraction

$$(33) \quad \rho_N : \mathcal{A} \longrightarrow \mathcal{A}_N.$$

Proposition 7.8 (Coefficientwise localisation). *Let α be a rational datum in $\mathcal{A}_N$, and put $P = (\mathcal{A}_N)_\alpha$. There is a canonical topological algebra isomorphism*

$$(34) \quad \mathcal{A}_\alpha \xrightarrow{\sim} \widehat{\bigoplus_\mu}^{c_0} P e_\mu.$$

It is natural for rational refinements represented in a finite head: if $P \rightarrow Q$ is the corresponding head restriction map, the induced restriction sends

$$(35) \quad \sum_\mu p_\mu e_\mu \mapsto \sum_\mu (p_\mu|_Q) e_\mu.$$

The same statement holds after increasing N .

Proof. Let

$$R = \mathcal{A}_N \langle T_1, \dots, T_m \rangle, \quad J = \overline{(gT_i - f_i)_i} \subset R, \quad Q = R/J.$$

Evaluation at $T_i = f_i/g$ identifies Q with the head rational localisation P .

The tail decomposition applies after adjoining the variables T_i . Apply the quotient map $R \rightarrow Q$ to each tail coefficient. Since the rational relations lie in the head, this gives a continuous homomorphism

$$\Phi : \mathcal{A}_\alpha \longrightarrow \widehat{\bigoplus_\mu}^{c_0} Q e_\mu.$$

Conversely, the canonical map from Q to $\mathcal{A}_\alpha$ may be applied to every coefficient. For a null family $(q_\mu)_\mu$ the resulting series

$$\sum_\mu q_\mu e_\mu$$

converges in $\mathcal{A}_\alpha$, and defines a continuous homomorphism Ψ in the other direction. The multiplication rule (32) shows that both maps are algebra homomorphisms.

The two composites are the identity on finite tail sums with polynomial head coefficients. These are dense on both sides, so continuity gives $\Psi\Phi = 1$ and $\Phi\Psi = 1$. Composing with $Q \cong P$ proves (34).

For a rational refinement in the head, the two evaluation maps commute with the head restriction map. The preceding construction is therefore coefficientwise natural and gives (35). Presentation independence and transitivity give the same conclusion after enlarging the head. $\square$

Corollary 7.9 (Finite-head presentation). *Every rational localisation of $\mathcal{A}$ is canonically isomorphic to a ring of the form*

$$(36) \quad \widehat{\bigoplus_\mu}^{c_0} P e_\mu,$$

where P is a rational localisation of some affinoid head $\mathcal{A}_N$. Under this identification the inclusions obtained by enlarging the head have norm-nonincreasing algebra retractions, and their union is dense.

Proof. Start with a rational datum in $\mathcal{A}$. Multiplying all entries by one power of ϖ does not change the rational subset or localisation, so assume that the entries lie in $\mathcal{A}_0$. Choose an integral Bezout relation

$$\varpi^\ell = \sum_j a_j d_j, \quad a_j \in \mathcal{A}_0.$$

By (23), for some N the entries d_j admit approximations $d'_j \in \mathcal{A}_N \cap \mathcal{A}_0$ satisfying $d'_j - d_j \in \varpi^{\ell+1} \mathcal{A}_0$. By lemma 7.7, the primed datum gives the same rational subset and the same

completed localisation. The primed integral Bezout relation produced in that lemma remains valid after applying the retraction ρ_N ; hence the primed datum is a rational datum already in $\mathcal{A}_N$. Now apply proposition 7.8.

For $M \geq N$, put $P_M = (\mathcal{A}_M)_\alpha$. The same proposition, with head M , identifies the localisation with the c_0 -sum over the variables of index $> M$. Coefficient projection gives a norm-nonincreasing retraction to P_M , and finite tail truncation proves that $\bigcup_M P_M$ is dense. $\square$

7.5. Sheafiness.

Theorem 7.10. *The Huber pair $(\mathcal{A}, \mathcal{A}^\circ)$ is sheafy.*

Proof. We first work with a finite rational cover $U = \bigcup_i U_i$. Approximate the datum for U and the data for all the U_i simultaneously in one sufficiently large finite head. The small perturbation lemma does not change the rational domains, and proposition 7.8 gives compatible identifications

$$\mathcal{O}_X(U) \cong \widehat{\bigoplus_\mu^{c_0} P e_\mu}, \quad \mathcal{O}_X(U_i) \cong \widehat{\bigoplus_\mu^{c_0} P_i e_\mu},$$

with coefficientwise restriction maps. Here P is a rational localisation of one affinoid head and the P_i are rational localisations of P . The domains defined by the P_i cover $\mathrm{Spa}(P, P^\circ)$: pull a point back along the split coefficient retraction to the tail algebra, use the original cover, and then apply the retraction again.

Let $(x_i)_i$ be a matching family and write $x_i = \sum_\mu x_{i,\mu} e_\mu$. Since the head P is affinoid and sheafy, the matching family $(x_{i,\mu})_i$ glues to a unique $q_\mu \in P$ for every μ . It remains to check that $(q_\mu)_\mu$ is a null family. Its restrictions to every P_i tend to zero because each x_i belongs to a c_0 -sum. The product restriction map

$$P \longrightarrow \prod_i P_i$$

is a topological embedding, so it reflects convergence to zero. Hence $q_\mu \rightarrow 0$, and $\sum_\mu q_\mu e_\mu$ is the required section over U . The same coefficientwise argument proves separation.

The image of the product restriction map for $\mathcal{A}$ is therefore the matching-family equaliser. This equaliser is closed, and the restriction map is continuous and injective. The nonarchimedean closed-range theorem over the Tate ring $\mathcal{A}$ now makes it a topological embedding. We have proved the topological sheaf condition on finite rational covers. The general rational-basis theorem upgrades this to the all-open structure presheaf, and its plus-ring independence gives the same conclusion for every ring of integral elements. In particular, $(\mathcal{A}, \mathcal{A}^\circ)$ is sheafy. $\square$

7.6. Reducedness of all rational localisations.

Lemma 7.11. *Let P be a reduced ring and let J be any index set. Define*

$$(37) \quad \mathcal{F}_J(P) = \prod_{\mu \in \mathbf{N}^{(J)}} P U^\mu,$$

with multiplication by coefficientwise finite convolution. Then $\mathcal{F}_J(P)$ is reduced.

Proof. Since P is reduced, the diagonal map

$$P \longrightarrow \prod_{\mathfrak{p} \in \mathrm{Spec}(P)} P/\mathfrak{p}$$

is injective. Applying formal power series coefficientwise, and then commuting formal power series with this product, gives an injection

$$\mathcal{F}_J(P) \longrightarrow \prod_{\mathfrak{p} \in \text{Spec}(P)} \mathcal{F}_J(P/\mathfrak{p}).$$

Each factor on the right is a domain, since its coefficient ring is a domain. The product is therefore reduced, and so is $\mathcal{F}_J(P)$. $\square$

Theorem 7.12. *Every finite iterated rational localisation of $\mathcal{A}$ is reduced.*

Proof. By transitivity of rational localisation, it is enough to treat one rational localisation. Use corollary 7.9 to write it as

$$(38) \quad E \cong \widehat{\bigoplus_{\mu} P e_{\mu}}^{c_0}, \quad P = (\mathcal{A}_N)_{\alpha}.$$

The affinoid algebra $\mathcal{A}_N$ is a domain by lemma 7.2. A rational localisation of a reduced affinoid algebra is reduced [4, Corollary 7.3.2/10]; see also [9, Remark 1.2.16]. Thus P is reduced.

The element W is a non-zero-divisor in $\mathcal{A}_N$. Affinoid rational localisation is algebraically flat [4, Corollary 7.3.2/6]; see also [10, Proposition 15.1.1, proof]. Tensoring $0 \rightarrow \mathcal{A}_N \xrightarrow{W} \mathcal{A}_N$ with P therefore shows that multiplication by W is injective on P . If $P = 0$, the conclusion is immediate, so assume otherwise.

Let $J = \{n > N\}$. Define

$$(39) \quad \Phi : E \longrightarrow \mathcal{F}_J(P), \quad \Phi \left(\sum_{\mu} x_{\mu} e_{\mu} \right) = \sum_{\mu} W^{\omega(\mu)} x_{\mu} U^{\mu}.$$

The multiplication rule (32) makes Φ multiplicative. If $\Phi(\sum x_{\mu} e_{\mu}) = 0$, coefficient comparison gives $W^{\omega(\mu)} x_{\mu} = 0$ for every μ . The W -regularity of P then gives $x_{\mu} = 0$ for every μ , so Φ is injective. By lemma 7.11, the target is reduced; hence E is reduced. $\square$

Proof of theorem 7.1. The uniform domain assertion is proposition 7.3, and nonnoetherianity is proposition 7.4. Sheafiness is theorem 7.10. Reducedness of all finite iterated rational localisations is theorem 7.12. Finally, propositions 7.5 and 7.6 identify the rational chart $|W| \leq |\varpi|$ as a nonuniform integral domain. $\square$

Remark 7.13 (Relation with the first construction). The finite-jet and weighted-parity examples use different mechanisms. In the finite-jet ring, a whole tail becomes infinitely ϖ -divisible and the bad chart acquires a nilpotent. In the weighted-parity ring, every rational calculation can be moved into a noetherian finite head and then carried coefficientwise through a c_0 -tail. Its bad chart remains a domain, and nonuniformity is witnessed by the unbounded family $(T_n)_{n \geq 1}$ of nonnilpotent power-bounded elements. The use of an infinite monomial support and of unbounded power-bounded directions is in the tradition of the examples of Buzzard–Verberkmoes and Mihara; the finite-head coefficientwise descent is the additional feature that makes sheafiness and reducedness after iterated rational localisation accessible here.

APPENDIX A. LEAN FORMALISATION AND VERIFICATION

This appendix records the formalisation of the finite-jet construction and the present state of the weighted-parity construction. The finite-jet source is AINTLIB commit `b007a4f3d`; the weighted-parity source is the later work-in-progress commit `090a28921` on the branch `wp/reduced-example` [1]. The latter contains the construction and proves uniformity, sheafiness,

nonnoetherianity, the domain property, and failure of stable uniformity. It does not yet prove, without unproved inputs, that every finite iterated rational localisation is reduced.

Neither pinned snapshot was available from the public AINTLIB repository at the time of writing. The interactive version of the paper therefore includes each declaration used below, extracted from the appropriate snapshot. The verification report records the Lean and mathlib versions, the build, and the axiom checks.

A.1. Sheafiness. The finite-jet proof first establishes a condition on finite rational covers. Its historical internal class is `IsSheafy`, in the `ValuationSpectrum` namespace and defined in `StructureSheaf.lean`.

```
class IsSheafy (A : Type u) [CommRing A] [TopologicalSpace A]
  [IsTopologicalRing A] [inst₁ : PlusSubring A]
  [inst₂ : IsHuberRing A] [T2Space A] [NonarchimedeanRing A]
  [let I : UniformSpace A :=
    IsTopologicalAddGroup.rightUniformSpace A; CompleteSpace A]
  [IsRingOfIntegralElements (A+)] : Prop where
embedding : ∀ (C : RationalCoveringData A), C.IsRational →
  Topology.IsEmbedding (productRestrictionSub A C)
gluing : ∀ (C : RationalCoveringData A), C.IsRational →
  ∀ (f : ∀ (D : ↑C.covers), presheafValue D.1),
  (∀ (D₁ D₂ : ↑C.covers) (D₃ : RationalLocData A)
    (h₃₁ : rationalOpen D₃.T D₃.s ⊆
      rationalOpen D₁.1.T D₁.1.s)
    (h₃₂ : rationalOpen D₃.T D₃.s ⊆
      rationalOpen D₂.1.T D₂.1.s),
    restrictionMap D₁.1 D₃ h₃₁ (f D₁) =
      restrictionMap D₂.1 D₃ h₃₂ (f D₂)) →
  ∃ x : presheafValue C.base, ∀ (D : ↑C.covers),
    restrictionMap C.base D.1 (C.hsubset D.1 D.2) x = f D
```

The two fields say that restriction to a finite rational cover is a topological embedding and that every matching family glues. Compatibility is written using common rational refinements; the exact-intersection theorem identifies this with the usual compatibility on pairwise intersections. Thus `IsSheafy A` is the topological sheaf condition on the rational basis.

For an arbitrary open subset $V \subseteq \text{Spa}(A, A^+)$, `limitSections V` is the projective limit of the sections over the rational subdomains contained in V . The corresponding all-open condition is `IsLimitSheaf`, from `SheafyPair.lean`.

```
structure IsLimitSheaf (A : Type u) [CommRing A] [TopologicalSpace A]
  [IsTopologicalRing A] [PlusSubring A] [IsHuberRing A]
  [HasLocLiftPowerBounded A] : Prop where
injective : ∀ {V : Opens ↑(Spa A A+)} {ι : Type u}
  {U : ι → Opens ↑(Spa A A+)}
  (hle : ∀ i, U i ≤ V)
  (⊆ : (V : Set ↑(Spa A A+)) ⊆ ⋃ i, (U i : Set ↑(Spa A A+)))
  {x y : ↑(limitSections V)},
  (∀ i, limitRestrict (hle i) x = limitRestrict (hle i) y) → x = y
glue : ∀ {V : Opens ↑(Spa A A+)} {ι : Type u}
  {U : ι → Opens ↑(Spa A A+)}
  (hle : ∀ i, U i ≤ V)
  (⊆ : (V : Set ↑(Spa A A+)) ⊆ ⋃ i, (U i : Set ↑(Spa A A+)))
  (s : ∀ i, ↑(limitSections (U i))),
  (∀ i j,
    limitRestrict (inf_le_left (a := U i) (b := U j)) (s i) =
```

```

    limitRestrict (inf_le_right (a := U i) (b := U j)) (s j)) →
    ∃ x : ↑(limitSections V), ∀ i, limitRestrict (hle i) x = s i
isEmbedding : ∀ {V : Opens ↑(Spa A A+)} {ι : Type u}
{U : ι → Opens ↑(Spa A A+)}
(hle : ∀ i, U i ≤ V)
(⊆ : (V : Set ↑(Spa A A+)) ⊆ ⋃ i, (U i : Set ↑(Spa A A+))),
Topology.IsEmbedding (limitRestrictProd hle)

```

The fields `injective` and `glue` are the usual sheaf axioms on arbitrary opens. The final field says that the resulting map to the matching-family equaliser is also a topological embedding. This is the extra condition needed for a sheaf of complete topological rings.

For complete Tate pairs the rational-basis and all-open predicates are equivalent:

```

theorem isSheafy_iff_isLimitSheaf :
  IsSheafy A ↔ IsLimitSheaf A :=
  ⟨fun _ => isLimitSheaf_of_isSheafy,
   isSheafy_of_isLimitSheaf⟩

```

The forward implication is the sheaf-on-a-rational-basis theorem, and the converse is restriction to rational covers. The structure-presheaf equivalence identifies `IsLimitSheaf A` with Wedhorn’s topological-ring formulation [18, Remark 8.20]. Mathlib’s ordinary categorical sheaf predicate is weaker because it does not record the topology.

For a complete Tate ring the following definitions fix a ring of integral elements and then remove the choices of plus ring and completion. They are in `SheafyRing.lean`.

```

def IsSheafyFor (Aplus : RingOfIntegralElements A) : Prop :=
  let I := Aplus.toPlusSubring
  have I : IsRingOfIntegralElements (A+ : Subring A) := Aplus.2
  have I : HasLocLiftPowerBounded A := hasLocLiftPowerBounded_faithful
  IsLimitSheaf A

def IsSheafyComplete : Prop :=
  ∀ Aplus : RingOfIntegralElements A, IsSheafyFor A Aplus

def IsSheafyTateRing : Prop :=
  ∀ (P : PairOfDefinition A)
  (Bplus : RingOfIntegralElements (CompletionModel A P)),
  IsSheafyFor (CompletionModel A P) Bplus

```

The predicate `IsSheafyFor A Aplus` says that the Huber pair (A, A^+) is sheafy. For complete Tate rings, the chosen-plus-ring equivalence shows that this is independent of the chosen ring of integral elements. The equivalences in `SheafyCompletionModel.lean` remove the choice of completion. Consequently `IsSheafyTateRing` is the analytic specialisation of Wedhorn’s Definition 8.26 [18, Definition 8.26].

A.2. Strong noetherianity. The definitions below are from `RestrictedPowerSeries.lean`.

```

def MvPowerSeries.IsRestricted {k : ℕ} {A : Type*}
[CommRing A] [TopologicalSpace A]
(f : MvPowerSeries (Fin k) A) : Prop :=
  Tendsto (fun s : Fin k →0 ℕ => MvPowerSeries.coeff s f)
  cofinite (nhds 0)

class IsStronglyNoetherian (A : Type*) [CommRing A]
[TopologicalSpace A] [NonarchimedeanRing A] : Prop where

```

```
isNoetherianRing_restricted : ∀ k : ℕ,
  IsNoetherianRing (restrictedMvPowerSeriesSubring k A)
```

The cofinite convergence condition is the usual condition that the coefficients tend to zero, so `restrictedMvPowerSeriesSubring k A` represents $A\langle T_1, \dots, T_k \rangle$. The second declaration therefore says that these rings are noetherian for every k . Since A is already complete, this is Wedhorn's definition of strong noetherianity [18, Proposition and Definition 6.36].

A.3. The finite-jet example. We finally record the definition of the example itself. In the following code, `PowerSeries.Restricted R 1` is the radius-one Tate algebra in one variable over R , `RestrictedLaurent K` is $K\langle W, W^{-1} \rangle$, and `DualNumber R` is $R[Q]/(Q^2)$. The source is `FJP/Over/JetRings.lean`.

```
abbrev L : Type u := RestrictedLaurent K

abbrev JetC : Type u := PowerSeries.Restricted (L K) (1 : ℝ)

abbrev JetB : Type u :=
  DualNumber (PowerSeries.Restricted K (1 : ℝ))

abbrev JetD : Type u := DualNumber (L K)

noncomputable def jetSupport : Subring (JetC K) where
  carrier := {f |
    qCoeff K 0 f ∈ nonnegSubring K ∧
    qCoeff K 1 f ∈ nonnegSubring K}
  zero_mem' := by
    constructor <=> rw [qCoeff_zero] <=> exact zero_mem _
  one_mem' := by
    constructor <=> rw [qCoeff_one]
    · rw [if_pos rfl]; exact one_mem _
    · rw [if_neg one_ne_zero]; exact zero_mem _
  add_mem' := fun {f g} hf hg => by
    constructor <=> rw [qCoeff_add]
    · exact add_mem hf.1 hg.1
    · exact add_mem hf.2 hg.2
  neg_mem' := fun {f} hf => by
    constructor <=> rw [qCoeff_neg]
    · exact neg_mem hf.1
    · exact neg_mem hf.2
  mul_mem' := fun {f g} hf hg => by
    constructor
    · rw [qCoeff_zero_mul]
      exact mul_mem hf.1 hg.1
    · rw [qCoeff_one_mul]
      exact add_mem (mul_mem hf.1 hg.2) (mul_mem hf.2 hg.1)

abbrev JetA : Type u := ↑(jetSupport K)
```

The predicate `nonnegSubring K` cuts out $K\langle W \rangle$ inside $K\langle W, W^{-1} \rangle$. Consequently the carrier condition says that the coefficients of Q^0 and Q^1 have no negative powers of W , while the coefficients of Q^n for $n \geq 2$ are unrestricted. Thus `JetA K` is

$$\left\{ f_0(W) + Qf_1(W) + Q^2h(W, W^{-1}, Q) \mid \begin{matrix} f_0, f_1 \in K\langle W \rangle \\ h \in L\langle Q \rangle \end{matrix} \right\},$$

which is the coefficient description of A in equation (5) after inverting ϖ .

The file also constructs the four maps in the Milnor square and proves `milnorRow_exact`: every compatible pair in `JetB K` and `JetC K` comes from a unique element of `JetA K`. This identifies `JetA K` with $B \times_D C$. The sheaf argument produces the finite-rational-cover condition displayed above:

```
include  $\varpi$  hK0 in
theorem isSheafy_JetA :
  ValuationSpectrum.IsSheafy (JetA K) where
  embedding := fun C hC =>
    productRestrictionSub_isEmbedding_JetA  $\varpi$  hK0 C hC
  gluing := fun C hC f hcompat =>
    gluing_JetA  $\varpi$  hK0 C hC f hcompat
```

Here ϖ is a `Uniformizer K`, and the remaining input is the noetherianity of `unitBall K`. This is the version of sheafiness proved directly for the example. The equivalences above give the all-open and choice-independent statements. The main theorems are collected in `SheafyEndpoints`, and include the following.

```
theorem finiteJet_isDomain : IsDomain (JetA K) :=
  inferInstance

theorem finiteJet_not_noetherian :
   $\neg$  IsNoetherianRing (JetA K) :=
  not_isNoetherianRing_JetA K

theorem finiteJet_isUniform_of_dvr
  [IsDiscreteValuationRing  $\mathcal{O}[K]$ ] :
  TopologicalRing.IsUniform (JetA K) :=
  finiteJet_isUniform K (Uniformizer.ofDVR K)

theorem finiteJet_not_stablyUniform_of_dvr
  [IsDiscreteValuationRing  $\mathcal{O}[K]$ ] :
   $\neg$  TopologicalRing.IsStablyUniform (JetA K) :=
  finiteJet_not_stablyUniform K (Uniformizer.ofDVR K)

theorem finiteJet_isSheafyFor_of_dvr
  [IsDiscreteValuationRing  $\mathcal{O}[K]$ ] :
  IsSheafyFor (JetA K) (finiteJetPlus K) :=
  finiteJet_isSheafyFor K (Uniformizer.ofDVR K)
  (isNoetherianRing_unitBall K)

theorem finiteJet_isSheafyTateRing_of_dvr
  [IsDiscreteValuationRing  $\mathcal{O}[K]$ ] :
  IsSheafyTateRing (JetA K) :=
  finiteJet_isSheafyTateRing K (Uniformizer.ofDVR K)
  (isNoetherianRing_unitBall K)
```

Here K is any complete nontrivially normed ultrametric field for which $\mathcal{O}[K] = \{x : \|x\| \leq 1\}$ is a discrete valuation ring. No local compactness, finiteness of the residue field, characteristic, or perfection assumption is used. The theorem `finiteJet_isSheafyFor_of_dvr` proves that (A, A°) is sheafy, whilst `finiteJet_isSheafyTateRing_of_dvr` removes the choices of completion and ring of integral elements. Despite its name, the latter does not quantify over finite Tate extensions of A , so it is not a strong sheafiness statement. The corresponding

all-open statement is

```
TopCat.Presheaf.IsSheafOfTopologicalRings
  (ValuationSpectrum.structurePresheaf (JetA K)),
```

The corresponding Lean theorem proves this statement directly.

A.4. Scope of the finite-jet formalisation. The paper’s A_0 is represented by the Gauss-norm unit ball. The theorem `isPowerBounded_JetA_iff` identifies this unit ball with the power-bounded elements, giving $A^\circ = A_0$. Completeness and the Tate structure are supplied by the corresponding instances. As in the weighted-parity development, the constant-series map from K is explicit and isometric but is not installed as an `Algebra K` instance.

For the bad chart, `chartEquiv` is an exported ring equivalence. The two continuity theorems make it a homeomorphism, and the canonical-map theorem identifies W and Q . It is not bundled as a `K-AlgEquiv`, which is why proposition 4.1 is stated for complete topological rings.

The conclusions of lemma 5.1 are formalised in the normed setting used for the example: E is one of B, C, D , T_E has its Gauss norm, $T_{E,0}$ is its unit ball, and the finite free terms have their sup norms. Positive-degree exactness is transported from the polynomial sequence by flat base change. Closedness in positive degree comes from exactness and continuity, whilst the degree-zero ideal is closed by the noetherian closed-ideal theorem. The closed-range theorem then gives controlled lifts and strictness; scaling those lifts gives the two displayed denominator inclusions.

The proof of lemma 5.2 in Lean uses these controlled lifts directly. Its localisation proof obtains the topological embedding from a quotient-norm estimate, and proves openness of $C_\alpha \rightarrow D_\alpha$ from the norm-preserving coefficientwise section. The components of proposition 5.3 are therefore separate declarations. Finally, the development proves the concrete finite-jet transfer `isSheafy_JetA`: it pushes finite rational covers to B, C, D , glues there, and uses the localised Milnor row and a closed-range theorem. The abstract statement lemma 6.1 for every strict Milnor square is not separately bundled in Lean.

The Koszul argument, the rational-chart calculation, and the transfer of sheafiness through the concrete Milnor square are contained respectively in the restricted-exactness file and the strictness and closed-image file, `FJP/Over/Chart.lean`, and `FJP/Over/SheafTransfer.lean`.

A.5. The weighted-parity example. The second development uses the same restricted-series infrastructure. Its ambient ring is the countable restricted Tate algebra $K\langle W, U_1, U_2, \dots \rangle$. The definitions `wpWeight`, `WPMem`, and `wpSupport` are the direct translation of (14)–(18). The relevant part of the definition is as follows.

```
def wpWeight (w : ℕ → ℕ) (t : ℕ →₀ ℕ) : ℕ :=
  ∑ n ∈ t.support,
    if (t n).mod 2 = 1 ∧ n ≠ 0 then w n else 0

def WPMem (w : ℕ → ℕ) (t : ℕ →₀ ℕ) : Prop :=
  wpWeight w t ≤ t 0

abbrev WPA : Type _ := ↑(wpSupport K w)

def idWeight : ℕ → ℕ := fun n => n

abbrev WPAid : Type _ := WPA K idWeight
```

The coordinate $t(0)$ is the exponent of W ; the other coordinates are the exponents of the U_n . Thus `WPMem idWeight t` says exactly that $a \geq \omega(\nu)$, and `wpSupport` requires the coefficients

outside this monoid to vanish. It follows that $\text{WPAid } K$ is the ring $\mathcal{A}$ of (18). The maps `constA` and `norm_constA` give the isometric embedding of K as the constant series. This map is explicit in the development, although it is not bundled as an `Algebra K` instance.

Suppose that K is complete and nontrivially normed and that $\mathcal{O}[K]$ is a DVR. The principal exported statements are:

```

theorem weightedParity_isUniform_of_dvr :
  IsUniform (WPAid K) :=
  weightedParity_isUniform (Uniformizer.ofDVR K)

theorem weightedParity_isDomain :
  IsDomain (WPAid K) :=
  inferInstance

theorem weightedParity_not_noetherian :
  ¬ IsNoetherianRing (WPAid K) :=
  not_isNoetherianRing_WPA K idWeight idWeight_pos

theorem weightedParity_powerBounded_eq_unitBall
  (w : Uniformizer K) :
  powerBoundedSubring (WPAid K) =
  (FiniteJet.unitBall (WPAid K) : Set (WPAid K)) :=
  powerBoundedSubring_eq_unitBall w

theorem weightedParity_isSheafyComplete_of_dvr :
  IsSheafyComplete (WPAid K) :=
  weightedParity_isSheafyComplete (Uniformizer.ofDVR K)
  (FiniteJetOver.isNoetherianRing_unitBall K)

theorem weightedParity_chart_isDomain
  (w : Uniformizer K)
  (hK₀ : IsNoetherianRing (FiniteJet.unitBall K)) :
  IsDomain (presheafValue
    (chartDatum (w := idWeight) w)) :=
  isDomain_chart w hK₀

theorem weightedParity_chart_not_isUniform
  (w : Uniformizer K)
  (hK₀ : IsNoetherianRing (FiniteJet.unitBall K)) :
  ¬ IsUniform (presheafValue
    (chartDatum (w := idWeight) w)) :=
  not_isUniform_chart idWeight_unbounded w hK₀

theorem weightedParity_not_stablyUniform_of_dvr :
  ¬ IsStablyUniform (WPAid K) :=
  weightedParity_not_stablyUniform (Uniformizer.ofDVR K)
  (FiniteJetOver.isNoetherianRing_unitBall K)

```

Here `IsSheafyComplete` is the predicate defined above: it quantifies over every ring of integral elements. In particular it proves the sheaf condition for $(\mathcal{A}, \mathcal{A}^\circ)$. The all-open theorem also records the corresponding statement directly for the structure presheaf. The datum `chartDatum` is $(W; \varpi)$, and `chartDatum_isRational` checks that it defines the chart $|W| \leq |\varpi|$. The two chart theorems then say precisely that its section ring is a domain but is not uniform.

The proof routes are the ones used in Section 7. Multiplicativity of the countable Gauss norm gives the domain and unit-ball statements. Finite-head localisation is proved by taking

the completed head quotient, applying its evaluation map coefficientwise to the tail, and constructing the inverse by summing a c_0 -family; it does not invoke the Koszul lemma. For sheafiness, one common head is chosen for the base datum and all members of a finite cover. Head sheafiness glues each coefficient, and the head restriction embedding shows that the resulting coefficients tend to zero. The final topological embedding is obtained from the closed matching equaliser by the Tate closed-range theorem. For the bad chart, the head quotient is $K\langle X \rangle$ and the tail algebra embeds in formal multivariable power series over this domain. The Lean small-perturbation theorem is the norm-unit-ball specialisation used in this argument; the more general formulation of lemma 7.7 is a paper-level statement.

These declarations, including the final failure of stable uniformity, have been checked to use only the usual logical axioms `propext`, `Classical.choice`, and `Quot.sound`. The current declarations asserting reducedness after every finite chain of rational localisations, on the other hand, still depend on `sorryAx`. Clause (3) of theorem 7.1 is therefore not counted as formalised. The branch also proves sheafiness for the shifted-weight models and constructs bicontinuous equivalences with the corresponding Tate extensions. It includes a declaration named `weightedParityStronglySheafy`, whose conclusion concerns the shifted-weight model. It does not yet state the transported conclusion directly for the literal Tate-extension type; no strong-sheafiness claim for that type is made here.

A.6. Mathematical and code provenance. The formal structure presheaf, rational-cover criterion, and the strongly-noetherian sheaf theorem follow Huber’s construction [8, Theorem 2.2 and Lemmas 2.3–2.6] and Wedhorn’s systematic formulation [18, Proposition and Definition 6.36, Remark 8.20, Definition 8.26, Theorem 8.28(b), Corollary 8.32, and Lemmas 8.33–8.34]. The passage between a fixed plus ring and all plus rings follows the standard rational-refinement argument recorded by Kedlaya [9, Lemma 1.6.8 and Remark 1.6.9]. Related strict Koszul statements appear in the work of Ben-Bassat–Kremnizer and Bambozzi–Kremnizer. In the formalisation, positive-degree exactness is proved by flat base change from the polynomial sequence; closedness and the nonarchimedean open mapping theorem [11, Lemma 3.1] give controlled lifts. Scaling those lifts yields the bounded-denominator criterion used by Buzzard–Verberkmoes [5, Lemma 2].

The development is written in Lean 4 [12] on top of `mathlib` [13]. The formalisation-specific files live in `AINTLIB` [1]. Its restricted power-series and restricted Laurent-series infrastructure includes Apache-2.0-licensed code adapted from William Coram’s PhD repository, with equivalence code attributed there to Bingyu Xia [6]. The vendored source headers and commit history retain those attributions. The finite-jet and weighted-parity definitions, localisation proofs, and sheaf endpoints discussed in this appendix are the `AINTLIB`-specific part of the development.

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